REPORT DOCUMENTATION PAGE

Form Approved
OMB No. 0704-0188

1. REPORT DATE *(DD-MM-YYYY)*
11/07/2019

2. REPORT TYPE
Research

3. DATES COVERED *(From - To)*

4. TITLE AND SUBTITLE
Classical Trajectory Monte Carlo Simulation of Plasma Fueling Using Magnetic Plasma Expulsion

5a. CONTRACT NUMBER

5b. GRANT NUMBER
NSF: PHY-1500427 and PHY-1803047;
DOE: PHY-1500427 and PHY-1803047

5c. PROGRAM ELEMENT NUMBER

6. AUTHOR(S)
Martinez, A.;
Ordonez, C.A.

5d. PROJECT NUMBER

5e. TASK NUMBER

5f. WORK UNIT NUMBER

7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES)
Department of Physics, University of North Texas, Denton, Texas 76203, USA

8. PERFORMING ORGANIZATION REPORT NUMBER

9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES)
National Science Foundation (NSF), Alexandria, VA, USA
Department of Energy (DOE), Washington, D.C., USA

10. SPONSOR/MONITOR'S ACRONYM(S)

11. SPONSOR/MONITOR'S REPORT NUMBER(S)

12. DISTRIBUTION/AVAILABILITY STATEMENT
DISTRIBUTION A. Approved for public release: distribution unlimited.

13. SUPPLEMENTARY NOTES

14. ABSTRACT
The possibility of fueling a magnetically confined plasma using particle sources located inside of the plasma is studied by computer simulation. Magnetic plasma expulsion [R. E. Phillips and C. A. Ordonez, Phys. Plasmas 25, 012508 (2018)] would serve to keep the magnetically confined plasma away from the particle sources without adversely affecting plasma confinement. The simulations show how charged particles can be injected into a plasma by using particle sources located directly between two current-carrying wires that create a magnetic expulsion field. Plasma fueling with the average energy of injected particles greater than the average energy of plasma particles may serve for heating the plasma. Also, plasma fueling with positive and negative particles injected at different rates may serve for changing the neutrality of the plasma. Conditions for plasma fueling are investigated using a classical trajectory Monte Carlo simulation. Two types of particle sources are considered, and the fraction of emitted particles that reach (and fuel) the magnetically confined plasma is evaluated for each.

15. SUBJECT TERMS
magnetic plasma expulsion, magnetic shielding, Monte Carlo methods, magnetic mirrors, leptons, electromagnetic coils, plasma heating, positron-electron plasma confinement,trap-based techniques, fusion energy, plasma physics

16. SECURITY CLASSIFICATION OF: a. REPORT	b. ABSTRACT	c. THIS PAGE	17. LIMITATION OF ABSTRACT	18. NUMBER OF PAGES	19a. NAME OF RESPONSIBLE PERSON
U	U	U	U	8	19b. TELEPHONE NUMBER *(Include area code)*

Classical trajectory Monte Carlo simulation of plasma fueling using magnetic plasma expulsion

Cite as: AIP Advances **9**, 075209 (2019); https://doi.org/10.1063/1.5110259
Submitted: 15 May 2019 . Accepted: 02 July 2019 . Published Online: 11 July 2019

A. Martinez, and C. A. Ordonez

Classical trajectory Monte Carlo simulation of plasma fueling using magnetic plasma expulsion

Cite as: AIP Advances 9, 075209 (2019); doi: 10.1063/1.5110259
Submitted: 15 May 2019 • Accepted: 2 July 2019 •
Published Online: 11 July 2019

View Online Export Citation CrossMark

A. Martinez and C. A. Ordonez[a)]

AFFILIATIONS

Department of Physics, University of North Texas, Denton, Texas 76203, USA

[a)]Author to whom correspondence should be addressed. Electronic mail: cao@unt.edu.

ABSTRACT

The possibility of fueling a magnetically confined plasma using particle sources located inside of the plasma is studied by computer simulation. Magnetic plasma expulsion [R. E. Phillips and C. A. Ordonez, Phys. Plasmas **25**, 012508 (2018)] would serve to keep the magnetically confined plasma away from the particle sources without adversely affecting plasma confinement. The simulations show how charged particles can be injected into a plasma by using particle sources located directly between two current-carrying wires that create a magnetic expulsion field. Plasma fueling with the average energy of injected particles greater than the average energy of plasma particles may serve for heating the plasma. Also, plasma fueling with positive and negative particles injected at different rates may serve for changing the neutrality of the plasma. Conditions for plasma fueling are investigated using a classical trajectory Monte Carlo simulation. Two types of particle sources are considered, and the fraction of emitted particles that reach (and fuel) the magnetically confined plasma is evaluated for each.

I. INTRODUCTION

Magnetic plasma expulsion is an artificial phenomenon in which a magnetic field is produced for keeping plasma away from a nonmagnetic material object that is inserted into a plasma. For example, magnetic plasma expulsion may serve to keep plasma away from support cables or other types of mechanical supports that hold plasma-immersed magnetic coils in place. Magnetic plasma expulsion, also known as magnetic shielding or guarding, was conceptualized long ago.[1] Along with the initial research that was reported back then, Ref. 2 has recently reported research on the phenomenon using particle-in-cell computer simulation. It should also be noted that a magnetic field configuration similar to that considered in Ref. 2 for magnetic plasma expulsion was also studied for controlling space charge neutralized ion beams.[3]

A Lockheed Martin group and a group in Australia have proposed a magnetic confinement fusion concept and an inertial electrostatic confinement fusion concept, respectively, that employ magnetic coils located inside a fusion-energy generating plasma system.[4,5] Such coils would produce magnetic fields for confining plasma and/or for keeping plasma away from the coils. However, the plasma-immersed magnetic coils must be held in place by some means to keep them from moving. Both concepts may benefit from employing magnetic plasma expulsion to help avoid the adverse effects associated with the presence of mechanical supports that pass through the plasma to the plasma-immersed magnetic coils.[2,4]

Investigations of pair plasmas may advance knowledge associated with various phenomena of current interest including magnetobound states of positronium and astrophysical jets.[6–10] The interaction of positrons and electrons has been demonstrated in Penning traps,[11] although magnetic confinement of a pair plasma consisting of electrons and positrons at equal densities and temperatures has not been reported. A compact superconducting levitated dipole trap and a magnetic mirror trap have been discussed for confining low-energy electron-positron plasmas,[12,13] and a recent report describes a cross-field transport mechanism for lossless injection of positrons into a magnetic dipole trap.[14] Experiments such as the Levitated Dipole Experiment and Ring Trap 1 have each successfully employed a single plasma-immersed superconducting magnetic coil to magnetically confine plasmas.[15,16] Alternative confinement concepts have been studied involving multiple plasma-immersed magnetic coils.[17,18] While a single magnetic coil that is immersed within a plasma can be magnetically levitated to avoid plasma loss to mechanical supports, magnetic levitation of multiple coils or long single solenoids may not be feasible.

Magnetic plasma expulsion may serve as an enabling technology for alternative plasma confinement approaches that make use of multiple coils or a long single solenoid immersed within the confined plasma. The magnetic field of an alternative plasma confinement approach that makes use of a long single solenoid immersed within a confined plasma is illustrated in Fig. 1. Such a field could be produced by two coaxial solenoids, such that a magnetic mirror is produced along magnetic field lines that enclose the immersed solenoid, with the lowest field strength inside of the immersed solenoid. Plasma particles are axially confined within the immersed solenoid by the magnetic mirror effect. However, particles within the loss cone can follow magnetic field lines that pass between the solenoids. Suppose that material structures are present between the immersed solenoid and the outer solenoid, as illustrated in Fig. 2. Magnetic plasma expulsion could serve to reduce the possibility that particle trajectories intersect the material structures. In Fig. 2, relatively small cylindrical structures connect the two coaxial solenoids, with the location of particle sources represented by a cylindrical structure located between two other cylindrical structures. The two other cylindrical structures could enclose mechanical supports, electrical connections, and two current-carrying wires that generate a magnetic expulsion field. A possible magnetic plasma expulsion field superimposed on a magnetic plasma confinement field is illustrated in Fig. 3. The magnetic plasma confinement field is approximated as being uniform. The magnetic plasma expulsion field distorts the total field in such a way that the field lines tend to go around the small cylindrical structures instead of through them. The alternative plasma confinement approach is describable as a magnetic mirror with closed field lines, as a result of incorporating a magnetic expulsion field.

The work presented here investigates by computer simulation the possibility of having particle sources located within a magnetic expulsion field. Possible particle sources include radioisotope encapsulated positron sources,[10] lithium-ion emitting alumino-silicate sources,[19] and more conventional electron and ion sources. Plasma fueling in combination with magnetic plasma expulsion represents a possible way to inject energetic charged particles directly into the interior of a magnetically confined plasma to control the plasma's density. Although plasma fueling alone is considered within the study presented here, plasma heating and plasma neutrality control are also possibilities. Plasma heating may occur when the average energy of injected particles is larger than the average energy of confined plasma particles. Plasma neutrality control may be possible by injecting positive and negative particles at different rates. The prospect of fueling a magnetically confined plasma using a particle

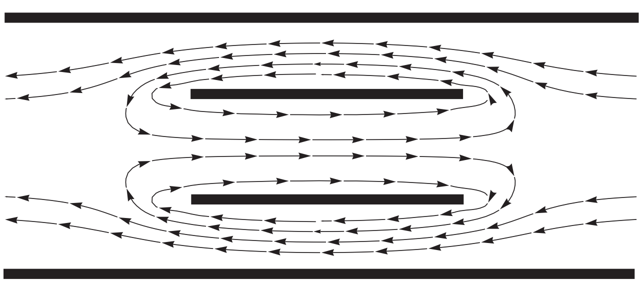

FIG. 1. Illustration of the cross section of two coaxial solenoids and an axisymmetric magnetic field that they produce. The shorter dark lines represent the immersed solenoid, and the longer dark lines represent the outer solenoid. The two solenoids produce a magnetic field that is weaker inside of the immersed solenoid than between the immersed solenoid and outer solenoid.

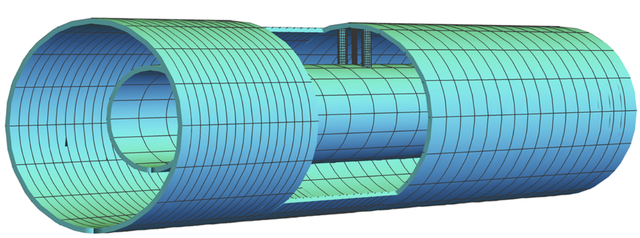

FIG. 2. Illustration of a dual-solenoid alternative magnetic plasma confinement approach. A cutout of the outer solenoid provides a partial view of an inner solenoid that would be immersed within the confined plasma. The cutout of the outer solenoid also provides a view of three small vertical cylindrical structures located between the two horizontal solenoids. The two outer vertical cylindrical structures enclose (1) current-carrying wires for producing a magnetic expulsion field, (2) mechanical supports that keep the immersed solenoid stationary, and (3) electrical connections that power the immersed solenoid. The middle vertical cylindrical structure represents the location of particle sources. Illustration is not to scale.

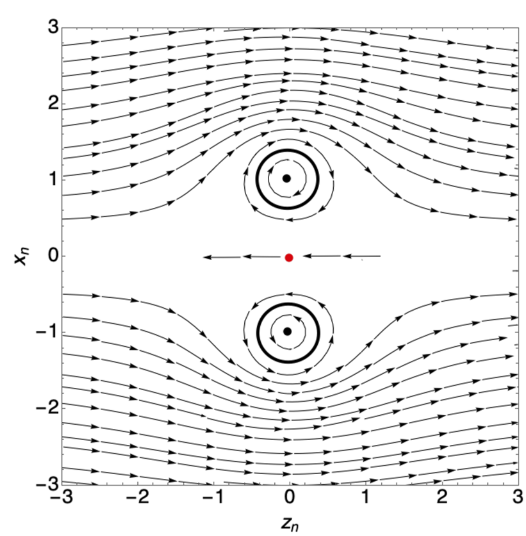

FIG. 3. Magnetic field distortion produced by superimposing the field generated by two parallel, current carrying wires on a uniform magnetic field. The uniform magnetic field represents the field produced by a magnetic plasma confinement approach, such as the field between the immersed solenoid and outer solenoid in Figs. 1 and 2. The hollow dark circles depict cylindrical structures that would enclose each of the two parallel wires, which are represented as black dots. The red dot represents the possible location of particle sources.

source located within a magnetic expulsion field is investigated with a classical trajectory Monte Carlo simulation. Section II provides a description of a magnetic field configuration used as a model for magnetic plasma expulsion and a description of normalized equations of motion that are evaluated numerically to simulate particle trajectories. Section III describes initial conditions, and Sec. IV reports simulation results. A discussion and concluding remarks are provided in Sec. V.

II. MAGNETIC FIELD MODEL AND NORMALIZED EQUATIONS OF MOTION

An investigation of plasma fueling is carried out by running a series of classical trajectory Monte Carlo simulations such that each simulation computes the trajectory of a particle subject to specified governing equations. An expression for the magnetic field is obtained by superimposing a uniform magnetic field and the magnetic fields produced by two infinitesimally thin, parallel, current-carrying wires. The uniform field represents a magnetic plasma confinement field, into which a particle source is to be inserted. The magnetic field produced by the two wires represents the magnetic plasma expulsion field. Any magnetic fields produced by other sources, such as the magnetically confined plasma or the particles from the particle source, are neglected. A Cartesian coordinate system is defined as illustrated in Fig. 3, with the z axis parallel to the uniform magnetic field, with the y axis parallel to each wire, and with the coordinate origin located midway between the two wires. The total magnetic field is written as

$$\boldsymbol{B}(x,y,z) = B_m\boldsymbol{\beta}(\frac{x}{S},\frac{y}{S},\frac{z}{S}) + B_0\hat{\boldsymbol{k}}, \tag{1}$$

where

$$\boldsymbol{\beta}(x_n,y_n,z_n) = [\frac{-z_n}{(x_n-1)^2+z_n^2},0,\frac{(x_n-1)}{(x_n-1)^2+z_n^2}] - [\frac{-z_n}{(x_n+1)^2+z_n^2},0,\frac{(x_n+1)}{(x_n+1)^2+z_n^2}]. \tag{2}$$

The product $B_m\boldsymbol{\beta}(x_n, y_n, z_n)$ describes the magnetic field generated by the two current carrying wires, which have currents flowing in opposite directions, and x_n, y_n, and z_n represent normalized Cartesian coordinates. $B_m = \mu_0 I/(2\pi S)$ is the magnetic field strength produced by a single wire at a distance S from the wire, I is the magnitude of the current carried by the wire, and μ_0 is the permeability of free space. The magnitude of the uniform magnetic field is denoted B_0, and $\hat{\boldsymbol{k}}$ is a unit vector that is parallel to the z axis. Figure 3 shows a magnetic field described by Eq. (1), which is not a function of the y coordinate.

The equations of motion that govern each particle's trajectory are normalized. All parameters are normalized by setting $m_n = q_n = K_n = S_n = 1$, where m_n is the normalized mass, q_n is the normalized charge and K_n is the normalized kinetic energy of a particle. S_n is the normalized distance between the y axis and each wire. Other parameters are normalized by writing them in terms of unnormalized parameters. The same symbol is used to denote both normalized and unnormalized parameters, except that each symbol that denotes a normalized parameter contains a subscript n. The normalized position is $\boldsymbol{x}_n = \boldsymbol{x}/S$, the normalized velocity is $\boldsymbol{v}_n = \boldsymbol{v}\sqrt{m/K}$, the normalized acceleration is $\boldsymbol{a}_n = \boldsymbol{a}mS/K$, the normalized time is $t_n = (t/S)\sqrt{K/m}$, and the normalized magnetic field is $\boldsymbol{B}_n = \boldsymbol{B}qS/\sqrt{mK}$. By solving for unnormalized variables and substituting them into classical equations of motion associated with the Lorentz force, $m\boldsymbol{a} = q\boldsymbol{v} \times \boldsymbol{B}$, three normalized equations of motion are obtained, as represented by the vector equation,

$$\boldsymbol{a}_n = \boldsymbol{v}_n \times \boldsymbol{B}_n. \tag{3}$$

In order to normalize the magnetic field given by Eq. (1), the uniform magnetic field strength B_0 is written in terms of the cyclotron radius r_0 for a particle of mass m, charge q, and kinetic energy K moving perpendicular to the field,

$$B_0 = \sqrt{\frac{2\,mK}{q^2 r_0^2}}. \tag{4}$$

The combined normalized expression for the uniform magnetic field superimposed with the field generated by the two wires is

$$\boldsymbol{B}_n(x_n,y_n,z_n) = \frac{\mathrm{sgn}(q)\sqrt{2}}{r_{0n}}[\beta_0\boldsymbol{\beta}(x_n,y_n,z_n) + \hat{\boldsymbol{k}}]. \tag{5}$$

Here, $\mathrm{sgn}(q) = q/|q|$ is the sign of charge, $r_{0n} = r_0/S$ is the normalized cyclotron radius, and

$$\beta_0 = \frac{B_m}{B_0} \tag{6}$$

is referred to as the field strength ratio.

III. INITIAL CONDITIONS

A classical trajectory Monte Carlo simulation is developed using a normalized description of classical particle motion, Eq. (3), with a normalized description of the total magnetic field, Eq. (5). A Monte Carlo approach is used to generate initial velocity components sampled from a velocity distribution associated with particles emitted from one of two particle sources that are considered. Each particle source is assumed to be located where indicated in Fig. 3, midway between the two straight wires that produce the magnetic expulsion field. The first particle source injects particles with a monoenergetic isotropic velocity distribution, and such particles are referred to collectively as a monoenergetic cloud. It should be emphasized that a classical trajectory Monte Carlo simulation does not account for plasma collective effects, because trajectories are simulated individually, and results only apply in the low density limit. The second particle source injects particles as a monoenergetic beam, where particles of equal kinetic energy all travel in the same direction. The first injection approach may serve as a model for a radioisotope source that emits positrons, and the second injection approach may serve as a model of a field emission source.

Particles tend to follow magnetic field lines in the vicinity of the current carrying wires and either immediately leave the region to fuel the magnetically confined plasma or remain orbiting one of the wires. The fraction F_0 of particles that reach $|z_n| = 5$ during a normalized time period $t_{n,max}$ are considered to fuel the confined plasma. Particle trajectories are simulated for a normalized time period, $t_{n,max} = 100/\sqrt{2}$. A particle traveling in a straight line would travel 100 times the distance between a wire and the y-axis during a normalized time period, $t_{n,max} = 100/\sqrt{2}$. Each evaluation of the value of F_0 is carried out by simulating 10,000 trajectories of positive particle [sgn$(q) = 1$].

Unless noted otherwise, all particles are approximated as originating from the same location at the coordinate origin,

$$x_n(0) = x_{0n} = 0, \tag{7}$$

$$y_n(0) = y_{0n} = 0, \tag{8}$$

$$z_n(0) = z_{0n} = 0. \tag{9}$$

However, for some simulations, nonzero values of x_{0n} and/or z_{0n} are used to assess the effect of a finite size for the particle source.

For a normalized kinetic energy, $K_n = 1$, non-relativistic particles with normalized mass $m_n = 1$ have a normalized speed, $v_{0n} = \sqrt{2}$. The normalized initial velocity components of a particle are written in terms of spherical coordinates as

$$v_{0nx} = \sqrt{2}\sin\theta_0\cos\phi_0, \tag{10}$$

$$v_{0ny} = \sqrt{2}\sin\theta_0\sin\phi_0, \tag{11}$$

$$v_{0nz} = \sqrt{2}\cos\theta_0, \tag{12}$$

where θ_0 and ϕ_0 are the initial polar and azimuthal angle coordinates, respectively.

The two injection approaches mentioned above are associated with different initial velocity distributions. For particle injection as a monoenergetic cloud, which has an isotropic velocity distribution with monoenergetic particles, θ_0 and ϕ_0 are sampled from a probability density function $f(\boldsymbol{v}_n)$. An isotropic velocity distribution for describing monoenergetic particles may be written in spherical coordinates as

$$f(v_{0n},\theta_0,\phi_0) = f_0\delta(v_{0n}-\sqrt{2})f_\theta(\theta_0)f_\phi(\phi_0). \tag{13}$$

Here, f_0 is a normalization constant, δ is the Dirac delta function, $f_\theta(\theta_0) = \sin\theta_0$, and $f_\phi(\phi_0) = 1$. The equation,

$$R_\theta = \frac{\int_{\theta_l}^{\theta_0}\sin\theta d\theta}{\int_{\theta_l}^{\theta_u}\sin\theta d\theta}, \tag{14}$$

is solved for the initial polar angle, with $\theta_l = 0$ and $\theta_u = \pi$, to obtain the sampling expression,

$$\theta_0 = \arccos(1-2R_\theta). \tag{15}$$

The sampling expression for the initial azimuthal angle is

$$\phi_0 = 2\pi R_\phi. \tag{16}$$

Here, R_θ and R_ϕ each have a uniform probability of being any random real number between 0 and 1, and each initial velocity vector is equally likely to be oriented in any direction.

For particle injection as a monoenergetic beam, all particles have the same kinetic energy and travel in the same direction. Two monoenergetic beam cases are considered, which are associated with what are referred to as a parallel monoenergetic beam and a perpendicular monoenergetic beam. In the parallel monoenergetic beam case, all particles are considered to initially travel parallel ($\theta_0 = 0$) to the z axis, and the following initial velocity components are used:

$$v_{0nx} = 0, \tag{17}$$

$$v_{0ny} = 0, \tag{18}$$

$$v_{0nz} = \sqrt{2}. \tag{19}$$

In the perpendicular monoenergetic beam case, all particles are considered to initially travel perpendicular to the z axis by using

$$v_{0nx} = 0, \tag{20}$$

$$v_{0ny} = \sqrt{2}, \tag{21}$$

$$v_{0nz} = 0. \tag{22}$$

IV. RESULTS

A. Particle injection as a monoenergetic cloud or as a parallel monoenergetic beam

Simulations with two different initial velocity distributions, one for a monoenergetic cloud and one for a monoenergetic beam, are conducted with all trajectories starting at the coordinate origin ($x_{0n} = y_{0n} = z_{0n} = 0$). Also, all particles in the monoenergetic beam are considered to initially travel in the z dimension ($v_{0nx} = v_{0ny} = 0$, $v_{0nz} = \sqrt{2}$). The fraction F_0 of particles that fuel the magnetically confined plasma is determined by varying r_{0n} for fixed values of β_0, with $\beta_0 = 1$, $\beta_0 = 2$, or $\beta_0 = 3$. It should be noted that the base case simulation of magnetic plasma expulsion in Ref. 2 was carried out with $\beta_0 = 1$. However, $\beta_0 = 2$ is also a particular value of interest. For $\beta_0 = 2$, the magnetic field produced by one wire at the location of the second wire is equal and opposite to the uniform field and results in zero net force acting on each current-carrying wire (in the infinite wire length approximation). Consequently, the value $\beta_0 = 2$ may approximately minimize the magnetic force each mechanical structure associated with a wire that passes through the plasma would have to withstand. For $\beta_0 = 1$, the magnetic field produced by one wire at the location of the second wire is half the strength of the uniform magnetic field, and, with $\beta_0 = 3$, a wire field strength that is 1.5 times stronger than the uniform field is produced at the other wire.

Figure 4 shows the fraction of simulated particles that fuel the confined plasma for the two initial velocity distributions that are considered. To interpret the results, it is helpful to solve Eq. (4) for the normalized cyclotron radius to obtain,

$$r_{0n} = \frac{\sqrt{2mK}}{B_0|q|S} = \frac{\beta_0\sqrt{2mK}}{B_m|q|S}. \tag{23}$$

Here, the equality on the right is written by using Eq. (6). For a given set of values for mass, kinetic energy, charge, wire separation distance, and field strength ratio β_0, an increasing value of the normalized cyclotron radius r_{0n} is associated with a simultaneous decrease in the parameters, B_0 and B_m, which characterize the strengths of the uniform magnetic field and the magnetic field produced by the wires, respectively. The results in Fig. 4 indicate that such a decrease results in a higher fraction of particles fueling the plasma when a monoenergetic cloud injection approach is used. Alternatively, for a

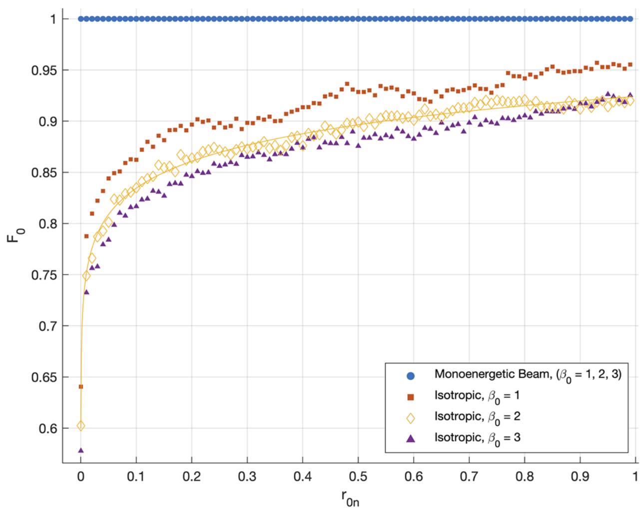

FIG. 4. Results for two particle injection approaches, an isotropic monoenergetic cloud and a parallel monoenergetic beam. All particles start at the coordinate origin, and monoenergetic beam particles initially move in the z dimension. For a given value of r_{0n}, the results for a monoenergetic beam are indistinguishable for each value of β_0, consisting of $\beta_0 = 1$, $\beta_0 = 2$, and $\beta_0 = 3$. The solid line is Eq. (24), which is a fit to the isotropic monoenergetic cloud results for $\beta_0 = 2$.

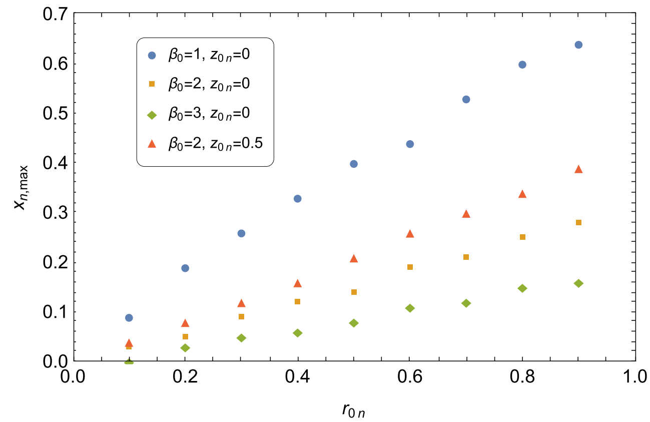

FIG. 5. Results for the parallel monoenergetic beam injection approach, with all particles initially moving in the z dimension. The largest values of x_{0n} for which all particles fuel the confined plasma $[F_0(0 \le x_{0n} \le x_{n,\max}) = 1]$ are plotted. The results are found by initially varying the value of x_{0n} in the range $0 \le x_{0n} \le 0.65$ in increments of 0.05 and using the initial results to vary the value of x_{0n} in a smaller range in increments of 0.01. Three sets of results for particles starting at $z_{0n} = 0.0$ (with $\beta_0 = 1$, $\beta_0 = 2$ and $\beta_0 = 3$), and one set of results for particles starting at $z_{0n} = 0.5$ (with $\beta_0 = 2$) are shown.

given value of β_0 with $1 \le \beta_0 \le 3$, the results in Fig. 4 indicate that the fraction of particles that fuel the plasma when a monoenergetic cloud injection approach is used increases with the mass and kinetic energy of the injected particles and decreases with the charge magnitude. Also shown in Fig. 4 is a plot of a fitted function obtained by applying the least squares method to the simulated data with $\beta_0 = 2$,

$$F_0 = 1.345 r_{0n}^{0.0299} - 0.421, \tag{24}$$

where r_{0n} is given by Eq. (23) with $\beta_0 = 2$.

For the monoenergetic beam injection approach, changes in r_{0n} and β_0 do not affect the fraction of particles that fuel the plasma, and the simulation results for $\beta_0 = 1$, $\beta_0 = 2$, and $\beta_0 = 3$ are indistinguishable. The result, $F_0 = 1$, occurs for all values of r_{0n} and β_0 considered. Such a result is understandable as an idealization, because each particle starts its trajectory at the coordinate origin with an initial velocity directed parallel to the z axis. The magnetic field has no x or y component along the z axis, and no magnetic force acts on a particle that travels along the z axis.

B. Parallel monoenergetic beam injection with finite width

Simulations with an initial velocity distribution associated with a monoenergetic beam injection approach are conducted with all particles considered to initially travel in the z dimension ($v_{0nx} = v_{0ny} = 0, v_{0nz} = \sqrt{2}$). The fraction F_0 of particles that fuel the magnetically confined plasma is determined by varying x_{0n} for given values of r_{0n}, β_0, and z_{0n}. It is found that, for given values of r_{0n}, β_0, and z_{0n}, there can exist a range of values that x_{0n} can have with $F_0 = 1$. The finding may be important, because particles that do not fuel the magnetically confined plasma are likely to bombard a material surface, resulting in undesirable particle-solid interactions. The results are presented by determining a maximum value $x_{n,\max}$ for which $F_0(0 \le x_{0n} \le x_{n,\max}) = 1$ for given values of r_{0n}, β_0, and z_{0n}. For determining conditions that satisfy $F_0(0 \le x_{0n} \le x_{n,\max}) = 1$, the value of x_{0n} is varied in the range $0 \le x_{0n} \le 0.65$. The results are shown in Fig. 5. The results indicate that, for a given value of β_0 and z_{0n}, there can exist an effective threshold value of r_{0n} above which $x_{n,\max} > 0$ occurs.

Equations (6) and (23) indicate that, for a given set of values for mass, kinetic energy, charge, wire separation distance, and normalized cyclotron radius r_{0n}, a decreasing value of the field strength ratio β_0 is a associated with a decrease in the parameter B_m, which characterizes the strength of the magnetic field produced by the wires. The results in Fig. 5 indicate that such a decrease results in a larger value of $x_{n,\max}$, when the value of r_{0n} is above the threshold value.

By way of example, the result $x_{n,\max} = 0.14$ is considered, which occurs for $r_{0n} = 0.5$, $\beta_0 = 2$ and $z_{0n} = 0$. The result indicates that a monoenergetic beam of charged particles that is emitted from a particle source at the $z = 0$ plane and that initially travels in the $\hat{\boldsymbol{k}}$ direction will fuel the magnetically confined plasma with $F_0 = 1$, provided that the width of the beam is not larger than $w_{\max} = 2x_{n,\max}S = 0.28S$ and the kinetic energy of each particle is larger than $K_{\min} = (r_{0n}B_0qS)^2/(2m) = (B_0qS)^2/(8m)$ with $\beta_0 = 2$. Because of the mass dependence, the kinetic energy of injected electrons or positrons must be much larger than that for singly charged ions, all other factors being equal, for the condition to be satisfied.

C. Perpendicular monoenergetic beam injection

Simulations with an initial velocity distribution associated with a monoenergetic beam injection approach are conducted with all particles considered to initially travel in the y dimension ($v_{0nx} = v_{0nz} = 0, v_{0ny} = \sqrt{2}$) and with all trajectories starting at the coordinate origin ($x_{0n} = y_{0n} = z_{0n} = 0$). The fraction F_0 of emitted particles that fuel the magnetically confined plasma is evaluated for each pair of values for r_{0n} and β_0 that were also used for the results in Fig. 5. The same result was found for each simulation, $F_0 = 0$. Similar to the monoenergetic beam injection approach with particle motion starting at the coordinate origin in the z dimension, the result with particle motion starting at the coordinate origin in the y dimension is understandable as an idealization. Each particle starts its trajectory at the coordinate origin with an initial velocity directed parallel to the y axis. The magnetic field only has a z component on the $z = 0$ plane. By directing a monoenergetic beam in the y direction with particles starting on the $z = 0$ plane, the particles continue to move perpendicular to the magnetic field, without a velocity component parallel to $\hat{\boldsymbol{k}}$ to move the particles out of the expulsion region. All particles are found to remain within the expulsion region and no plasma fueling is found to occur for all combinations of r_{0n} and β_0 considered.

The two monoenergetic beam injection approaches, one involving initial particle motion in the z dimension and the other involving initial particle motion in the y dimension, with both having particles start at the coordinate origin, represent two extremes in the possible results for the fraction of particles that fuel the confined plasma, $F_0 = 1$ and $F_0 = 0$, respectively. To see the transition from one result to the other, simulations with an initial velocity distribution associated with a monoenergetic cloud injection approach are conducted with all trajectories starting at the coordinate origin ($x_{0n} = y_{0n} = z_{0n} = 0$) and with $r_{0n} = 0.5$ and $\beta_0 = 2$. Figures 6 and 7 show scatter plots of pairs of random numbers, R_θ and R_ϕ, which are used in the sampling expressions, Eqs. (15) and (16), to determine the direction of the initial velocity of each particle. Figure 6 shows a scatter plot of pairs of (R_θ, R_ϕ) values for particles that fuel the plasma. For a situation in which every emitted particle reaches $|z| = 5$, each dot in Fig. 6 would be equally likely to have been located anywhere within a square defined by $0 \le R_\theta \le 1$ and $0 \le R_\phi \le 1$.

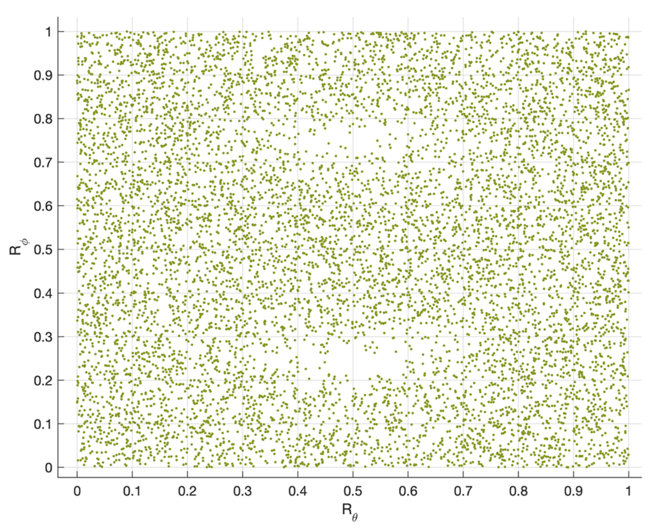

FIG. 6. Results for the isotropic monoenergetic cloud injection approach. All particles start at the coordinate origin, and the simulations are run with $r_{0n} = 0.5$ and $\beta_0 = 2$. Each dot of the scatter plot indicates the sampled values of R_θ and R_ϕ of particles that fuel the plasma.

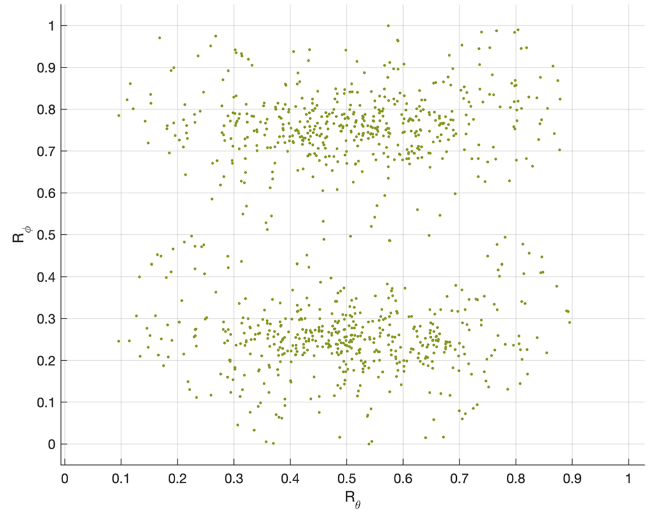

FIG. 7. Same as Fig. 6 except that a scatter plot of the sampled values of R_θ and R_ϕ of particles that do not fuel the plasma is shown.

However, according to Fig. 6, there are two regions where the plot tends to be devoid of dots, one region centered approximately at ($R_\theta = 0.5$, $R_\phi = 0.25$) and the other centered approximately at ($R_\theta = 0.5$, $R_\phi = 0.75$). Substitution into Eqs. (15) and (16) indicates that the corresponding initial angle coordinates are ($\theta_0 = \pi/2$, $\phi_0 = \pi/2$) and ($\theta_0 = \pi/2$, $\phi_0 = 3\pi/2$), respectively. Substitution into Eqs. (10) to (12) indicates that the corresponding initial Cartesian velocity components are ($v_{0nx} = 0$, $v_{0ny} = \sqrt{2}$, $v_{0nz} = 0$) and ($v_{0nx} = 0$, $v_{0ny} = -\sqrt{2}$, $v_{0nz} = 0$), respectively. Thus, the two regions in Fig. 6 where the plot tends to be devoid of dots are approximately centered at (R_θ, R_ϕ) values associated with particles that start trajectories at the coordinate origin with an initial velocity directed parallel or antiparallel to the y axis.

Figure 7 shows a scatter plot of pairs of (R_θ, R_ϕ) values for particles that do not fuel the plasma. According to Fig. 7, there are two regions where the plot tends to be devoid of dots for all values of R_ϕ, one region in the vicinity of $R_\theta = 0$ and the other in the vicinity of $R_\theta = 1$. Substitution into Eqs. (15) and (16) indicates that the corresponding initial polar angle coordinates are ($\theta_0 = 0$) and ($\theta_0 = \pi$), respectively. Substitution into Eqs. (10) to (12) indicates that the corresponding initial Cartesian velocity components are ($v_{0nx} = 0$, $v_{0ny} = 0$, $v_{0nz} = \sqrt{2}$) and ($v_{0nx} = 0$, $v_{0ny} = 0$, $v_{0nz} = -\sqrt{2}$), respectively. Thus, the two regions in Fig. 7 where the plot tends to be devoid of dots are approximately centered at R_θ values associated with particles that start trajectories at the coordinate origin with an initial velocity directed parallel or antiparallel to the z axis.

V. DISCUSSION AND CONCLUDING REMARKS

The density, temperature, and purity of a magnetically confined plasma can be significantly affected by inserting a material object into the plasma. Recent computer simulations support the possibility of keeping magnetically confined plasma out of a chosen region under certain conditions as a result of an artificial phenomenon referred to as magnetic plasma expulsion.[2] The artificial phenomenon may have a number of applications. For example, as mentioned in Ref. 2, an enhancement of how certain plasmas are magnetically confined may be possible by inserting particle sources that control both plasma neutrality and how the electric potential varies within the magnetically confined plasma. Also, the development of alternative magnetic plasma confinement approaches may be possible by inserting electromagnets that produce a portion of the magnetic field that confines the plasma. For both examples, magnetic plasma expulsion could serve to produce a magnetic field for keeping plasma away from nonmagnetic material objects that are inserted into the plasma, such as charged particle sources or mechanical support cables.

In the work presented here, an approach for fueling a magnetically confined plasma with particle sources located inside of the plasma has been studied with a classical trajectory Monte Carlo simulation. The magnetic field produced by two current-carrying wires is used as a model to represent the magnetic plasma expulsion field. The magnetic plasma expulsion field distorts the total field in such a way that the field lines tend to go around a chosen region instead of through the region. Normalized governing equations of motion are used such that the results depend on two parameters referred to as the normalized cyclotron radius r_{0n} and the magnetic field ratio β_0. The value $\beta_0 = 2$ is of particular interest, because it may approximately minimize the magnetic force on each mechanical structure associated with each current-carrying wire that passes through the magnetically confined plasma. Two types of particle sources were considered, and the fraction of emitted particles that reach and fuel the magnetically confined plasma was evaluated for each under various conditions. The particles from one particle source had a monoenergetic isotropic velocity distribution and the particles from the other particle source had a monoenergetic beam velocity distribution. The results indicate that the monoenergetic beam injection approach is the most promising, because all emitted particle were found to reach the magnetically confined plasma under certain conditions. A particle source that injects particles at a location directly between two current-carrying wires with an initial velocity parallel to that of the magnetic field that magnetically confines plasma may maximize the fraction of injected particles that reach the plasma.

ACKNOWLEDGMENTS

The authors would like to thank Dr. Ryan Phillips for helpful comments, suggestions, and discussions. Computational resources were provided by UNT's High Performance Computing Services. This material is based upon work supported by the National Science Foundation under Grant Nos. PHY-1500427 and PHY-1803047 and by the Department of Energy under Grant No. DE-FG02-06ER54883.

REFERENCES

[1] B. Lehnert, "Plasma confinement in presence of magnetically shielded supports," Plasma Physics **17**, 501–524 (1975) and references therein.

[2] R. E. Phillips and C. A. Ordonez, "Magnetic plasma expulsion," Physics of Plasmas **25**, 012508 (2018).

[3] R. E. Phillips and C. A. Ordonez, "Magnetic control of a charge-neutralized ion beam," Physics Procedia **66**, 148–155 (2015).

[4] J. Heinrich, F. H. Ebersohn, G. Font, M. L. Garrett, J. Guyton, A. Qerushi, and T. J. McGuire, "Permanent magnet support shielding for T4B," Bulletin of the American Physical Society **63**, CP11.00143 (2018).

[5] J. Hedditch, R. Bowden-Reid, and J. Khachan, "Fusion energy in an inertial electrostatic confinement device using a magnetically shielded grid," Physics of Plasmas **22**, 102705 (2015).

[6] J. R. Correa and C. A. Ordonez, "Magnetobound positronium and protonium," Physics of Plasmas **21**, 082115 (2014).

[7] F. F. Aguirre and C. A. Ordonez, "Giant cross-magnetic-field steps due to binary collisions between pair particles," Physical Review E **91**, 033103 (2015).

[8] F. F. Aguirre and C. A. Ordonez, "Giant increase in cross-magnetic-field transport rate as an electron-positron plasma cools," Physics of Plasmas **24**, 104502 (2017).

[9] J. F. C. Wardle, D. C. Homan, R. Ojha, and D. H. Roberts, "Electron–positron jets associated with the quasar 3C279," Nature **395**, 457–461 (1998).

[10] J. R. Danielson, D. H. E. Dubin, R. G. Greaves, and C. M. Surko, "Plasma and trap-based techniques for science with positrons," Reviews of Modern Physics **87**, 247–306 (2015).

[11] R. G. Greaves and C. M. Surko, "An electron-positron beam-plasma experiment," Physical Review Letters **75**, 3846–3849 (1995).

[12] M. R. Stoneking, H. Saitoh, M. Singer, E. V. Stenson, J. Horn-Stanja, T. S. Pedersen, S. Nißl, U. Hergenhahn, N. Yanagi, C. Hugenschmidt, M. Dickmann, J. R. Danielson, and C. M. Surko, "Toward a compact levitated superconducting dipole for positron-electron plasma confinement," AIP Conference Proceedings **1928**, 020015 (2018).

[13] H. Higaki, C. Kaga, K. Fukushima, H. Okamoto, Y. Nagata, Y. Kanai, and Y. Yamazaki, "Simultaneous confinement of low-energy electrons and positrons in a compact magnetic mirror trap," New Journal of Physics **19**, 023016 (2017).

[14] E. V. Stenson, S. Nißl, U. Hergenhahn, J. Horn-Stanja, M. Singer, H. Saitoh, T. S. Pedersen, J. R. Danielson, M. R. Stoneking, M. Dickmann, and C. Hugenschmidt, "Lossless positron injection into a magnetic dipole trap," Physical Review Letters **121**, 235005 (2018).

[15] D. T. Garnier, A. K. Hansen, J. Kesner, M. E. Mauel, P. C. Michael, J. V. Minervini, A. Radovinsky, A. Zhukovsky, A. Boxer, J. L. Ellsworth, I. Karim, and E. E. Ortiz, "Design and initial operation of the LDX facility," Fusion Engineering and Design **81**, 2371–2380 (2006).

[16] H. Saitoh, Z. Yoshida, J. Morikawa, Y. Yano, H. Hayashi, T. Mizushima, Y. Kawai, M. Kobayashi, and H. Mikami, "Confinement of electron plasma by levitating dipole magnet," Physics of Plasmas **17**, 112111 (2010).

[17] R. A. Lane and C. A. Ordonez, "Classical trajectory Monte Carlo simulations of particle confinement using dual levitated coils," AIP Advances **4**, 077117 (2014).

[18] J. D. Wofford and C. A. Ordonez, "Dual levitated coils for antihydrogen production," AIP Conference Proceedings **1525**, 106–110 (2013).

[19] P. K. Roy, W. G. Greenway, and J. W. Kwan, "Source fabrication and lifetime for Li^+ ion beams extracted from alumino-silicate sources," Review of Scientific Instruments **83**, 043303 (2012).